Vectors I: Homework 2

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Scalar product: properties, angles, perpendicularity, projections, ratios. Q1 True/False from the Basics Worksheet (Scalar Product; statements (e) and (p) omitted). Q2–Q7 are past-paper selections (dot product / ratio only). Total: 42 marks.

Questions

1

For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$.	
(b)	$\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$.	
(c)	$\mathbf{i}\cdot\mathbf{j}=0$.	
(d)	$\mathbf{a}\cdot\mathbf{b}$ is a vector.	
(f)	If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$.	
(g)	$(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$.	
(h)	$(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector.	
(i)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$, where θ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$.	
(j)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}$.	
(k)	If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.	
(l)	$\lvert\mathbf{a}\cdot\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.	
(m)	$\mathbf{a}\cdot\mathbf{b}\le\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.	
(n)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle θ between $\mathbf{a}$ and $\mathbf{b}$ is acute or zero ($0^\circ\le\theta<90^\circ$).	
(o)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle θ between $\mathbf{a}$ and $\mathbf{b}$ is obtuse or 180° ($90^\circ<\theta\le180^\circ$).	
(q)	$\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar.	
(r)	$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+\lvert\mathbf{b}\rvert^2$.	
(s)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel.	
(t)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$ also holds if θ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$.	
(u)	Let θ be the angle ABC. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=\lvert\overrightarrow{BA}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$.	
(v)	The length of projection of $\mathbf{a}$ onto $\mathbf{b}$ is equal to the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.	
(w)	If $\lvert\mathbf{a}\rvert=\lvert\mathbf{b}\rvert$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$.	
(x)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$.	
(y)	Let θ be the angle ABC. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$.	
(z)	Let θ be the angle CBD. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$.	

2 [EJC 2024 Promo Q5]

$\mathbf{a}$ and $\mathbf{b}$ are vectors such that $|\mathbf{a}|=\sqrt{2}$ and $|\mathbf{b}|=\sqrt{6}$. The angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{5\pi}{6}$.

(a) Find the exact length of projection of $\mathbf{a}$ onto $\mathbf{b}$. **[2]**

(b) By considering $(3\mathbf{a}+2\mathbf{b})\cdot(3\mathbf{a}+2\mathbf{b})$, find the exact value of $|3\mathbf{a}+2\mathbf{b}|$. **[4]**

3 [SAJC 2024 Promo Q9(a)]

Relative to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel. It is given that $|\mathbf{a}|=9$, $|\mathbf{b}|=1$ and $|2\mathbf{a}+\mathbf{b}|=2\sqrt{74}$.

(i) By considering $(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$. **[3]**

(ii) Give a geometrical meaning of $|\mathbf{a}\cdot\mathbf{b}|$. **[1]**

(iii) The points P, Q and R have position vectors $7\mathbf{a}-5\mathbf{b}$, $6\mathbf{a}+5\mathbf{b}$ and $9\mathbf{a}+\lambda\mathbf{b}$ respectively. Given that these three points are collinear, find the value of λ. **[3]**

4 [CJC 2022 Promo Q9(i),(ii)]

Relative to the origin O, the position vectors of points A, B and C are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where $\mathbf{b}$ is a unit vector. It is given that $\mathbf{b}$ and $\mathbf{b}-\mathbf{a}$ are perpendicular and C lies on AB such that $AC:CB=3:1$.

(i) Show that $|\mathbf{a}|=\sec\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. **[3]**

By expressing $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$,

(ii) find the value of $|\mathbf{c}\cdot\mathbf{b}|$ and state the geometrical interpretation of $|\mathbf{c}\cdot\mathbf{b}|$. **[4]**

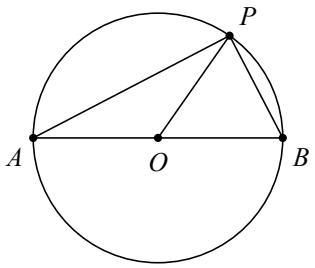
5 [SAJC 2025 Promo Q8(i)]

Relative to the origin O, two points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. It is given that $|\mathbf{a}|=2$, $|\mathbf{b}|=1$ and $|3\mathbf{a}-2\mathbf{b}|=\sqrt{37}$.

(i) By considering the scalar product $(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=\frac{1}{4}$, and give the geometrical meaning of $|\mathbf{a}\cdot\hat{\mathbf{b}}|$. **[4]**

6 [EJC 2025 Prelim P1 Q6]

There are three distinct points A, B and P. The point P lies on the circle with centre O and diameter AB. Relative to the origin O, the position vectors of points A and P are **a** and **p** respectively.



(a) By expressing $\overrightarrow{AP}$ and $\overrightarrow{BP}$ in terms of **a** and **p**, show that $\overrightarrow{AP}\cdot\overrightarrow{BP}=0$. **[3]**

The point C with position vector **c** lies on AB produced such that $AC:AB$ is $\lambda:1$.

(b) Find **c** in terms of λ and **a**. **[2]**

(c) It is given that PC is a tangent to the circle and that angle AOP is $120°$. Using a suitable scalar product, find the value of λ. **[4]**

7 [NJC 2025 Prelim P2 Q3]

With respect to an origin O, points P and Q have variable position vectors **p** and **q** respectively, given by

$$\mathbf{p}=(\cos t)\,\mathbf{i}+(\sin t)\,\mathbf{j}-\mathbf{k}, \qquad \mathbf{q}=(\cos 2t)\,\mathbf{i}-(\sin 2t)\,\mathbf{j}+\alpha\mathbf{k},$$

where α is a real parameter and $0<t<\pi$.

(i) Find the exact values of t and α if O, P and Q are collinear. **[5]**

It is given that $\alpha=0.5$.

(ii) Show that $\mathbf{p}\cdot\mathbf{q}=\cos 3t-0.5$. Find the greatest length of projection of **q** onto **p**. **[4]**

Answers

1 **(a)** T **(b)** T **(c)** T **(d)** F **(f)** T **(g)** T **(h)** T **(i)** T **(j)** F **(k)** F **(l)** F **(m)** T **(n)** T **(o)** T **(q)** F **(r)** F **(s)** T **(t)** T **(u)** T **(v)** F **(w)** T **(x)** F **(y)** F **(z)** F

2 **(a)** $\frac{\sqrt{6}}{2}$

(b) $\sqrt{6}$

3 **(i)** Shown.

(ii) Since $|\mathbf{b}|=1$, $|\mathbf{a}\cdot\mathbf{b}|$ is the length of the projection of **a** onto **b**.

(iii) $\lambda=-25$

4 **(i)** Shown.

(ii) $|\mathbf{c}\cdot\mathbf{b}|=1$; $|\mathbf{c}\cdot\mathbf{b}|$ is the length of the projection of **c** onto **b**.

5 **(i)** Shown. $|\mathbf{a}\cdot\hat{\mathbf{b}}|$ is the length of the projection of **a** onto **b**.

6 **(a)** Shown.

(b) $\mathbf{c}=(1-2\lambda)\mathbf{a}$

(c) $\lambda=\frac{3}{2}$

7 **(i)** $t=\frac{\pi}{3}$, $\alpha=1$ or $t=\frac{2\pi}{3}$, $\alpha=-1$

(ii) Shown. Greatest length $=\frac{3}{2\sqrt{2}}$ $(=1.06$ (3 s.f.))

END OF WORKSHEET